

Physics Project Portfolio

Dylan Miller | AP Physics 1, 2, and C: Mechanics | 2024-2026



[View Launch Video](#)

T-Shirt Cannon

Collaborated to design and develop a fully functional T-shirt cannon capable of safely launching shirts at a high-school pep rally with 500+ students present.

Tested 5+ air-pressure levels and 10 launch angles, measuring launch distance and accuracy to identify the most effective firing conditions.

Analyzed 5+ T-shirt folding methods and compared their effects on aerodynamics and travel distance.

Directed the initial fabrication process by applying experimental data to refine the mechanism and improve repeatability, accuracy, and overall performance.

Catapult

Evaluated 3 arm lengths, 5+ spring tensions, and 10 release angles to determine their effects on launch distance and precision.

Tested launches across 10 target distances successfully launching individual candy pieces, adjusting mechanical parameters based on measured results.

Challenged by the precise distance requirements and minimal margin for error, requiring careful design adjustments and iterative testing.

Independently led the mechanical development and construction of the catapult while teammates focused more on sketching and data collection.



[View Launch Video](#)



[View Video](#)

Rube Goldberg Machine

Designed and built a multi-stage Rube Goldberg machine incorporating 7+ energy transformations and ending with a successful balloon pop.

Integrated kinetic, potential, gravitational, chemical, and other forms of energy across sequential mechanisms.

Applied concepts from previous physics projects to coordinate multiple mechanisms into one reliable sequence.

Personally led the mechanical design of the final stages, specializing in using gravitational and kinetic energy to drive the balloon-popping mechanism.

Project Development Overview

DESIGN

- Developed mechanical designs from initial concepts into functional prototypes.
- Applied physics principles to determine how different design variables would affect performance.
- Tracked data to improve precision and functionality.

LEADERSHIP

- Led roles in developing concepts and directing the team's approach.
- Contributed key ideas while coordinating with teammates responsible for testing, sketches, and data collection.
- Translated ideas into physical components.

ITERATION

- Conducted controlled testing by altering variables such as pressure, spring tension, arm length, release angle, and more.
- Recorded results and used experimental data to make adjustments.
- Worked through tight margins of error and unexpected results.

APPLICATION

- Employed diverse concepts including various forms of energy.
- Analyzed projectile motion, physical transformations, and mechanical systems.
- Connected theoretical physics principles to hands-on engineering challenges.

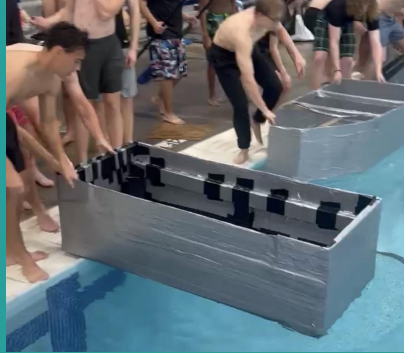
Other Projects

Toothpick Tower



Applied optimized construction techniques to build a 12-inch tower capable of supporting 385 pounds. Discovered through testing that triangular structures provided improved weight distribution and resistance to external forces, resulting in a stronger and more stable design.

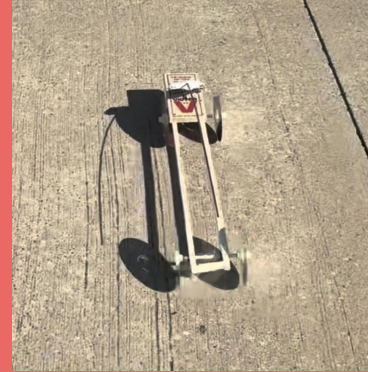
Cardboard Boat



Constructed a functional boat using only cardboard and duct tape that safely supported and fit three high schoolers. Designed for structural integrity and water resistance, the boat remained intact, dry, and afloat throughout the entire allotted race.

[View Video](#)

Mousetrap Car

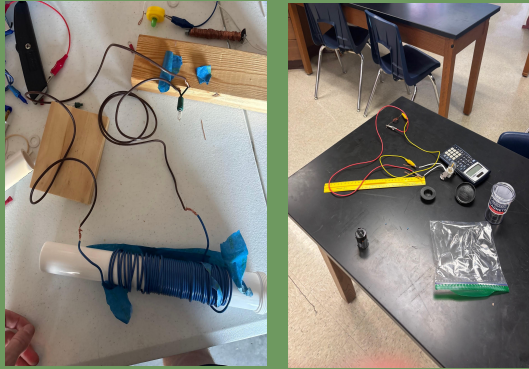


Utilized trial and error to determine how various aspects influenced the car's performance. Evaluated the weight and power of both mouse and rat traps to select the optimal mechanism; ultimately, designing a car capable of traveling 40ft.

[View Video](#)

Other Projects

Electric Generator



Designed and constructed a functional electric generator using electromagnetic fields generated by wire coils to induce motion. Overcame significant design and troubleshooting challenges with limited feedback, ultimately becoming one of the few groups to successfully produce a working generator.

Hot Air Balloon



Designed and constructed a hot-air balloon using plastic bags, acetone, and cotton balls capable of sustaining 5 paperclips in flight. Through trial and error, determined the optimal balance between bag weight, heat resistance, and fuel quantity to achieve flight.

Bolder Adventure Park



Collaborated on a comprehensive Physics I review project centered around Boulder Adventure Park, applying concepts from throughout the course to real-world scenarios. After completing the problem set, experienced the activities firsthand, connecting application to the packet problems.

[View Video](#)